

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.379969405	delta=+0.0283718
ratio=1.0120649		
CoG x [m]	-0.002024708562 -> -0.002397240376	delta=-0.000372532
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.001062500928	delta=+0.00109303
ratio=-		
CoG z [m]	0.009509749599 -> 0.01098342169	delta=+0.00147367
ratio=-		
rotor effectiveness 1	1 -> 0.8986308179	delta=-0.101369 ratio=0.89863082
rotor effectiveness 2	1 -> 0.9194583022	delta=-0.0805417 ratio=0.9194583
rotor effectiveness 3	1 -> 0.8874578635	delta=-0.112542 ratio=0.88745786
rotor effectiveness 4	1 -> 0.888999914	delta=-0.111 ratio=0.88899991
command lag [s]	0 -> 0.03819661874	delta=+0.0381966 ratio=-

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06500007669, -7.659918685e-05, 7.066962573e-06]
[-7.659918685e-05, 0.06530440782, 9.867924423e-05]
[ 7.066962573e-06, 9.867924423e-05, 0.1289838243]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06498187064	delta=+2.92255e-05
ratio=1.00045		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06532246019	delta=+0.000322393
ratio=1.0049599		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289839779	delta=-8.13839e-06
ratio=0.99993691		

Optimization

joint residual-wrench accumulated squared loss: 4.2420024372
Gaussian-prior cost: 2.26877781182
selected strict-ZOH lag [s]: 0.038196618742

Bag failure_1_single

reconstruction position RMSE [m]: 0.000652642996927
reconstruction position component RMSE [m]: [0.00034 0.000324 0.000453]
reconstruction orientation RMSE [deg]: 0.596167087291
gyro consistency RMSE xyz [rad/s]: [0.041104 0.085835 0.02655]
specific-force consistency RMSE xyz [m/s^2]: [0.953998 0.163222 0.343061]
inferred external force RMS xyz [N]: [0.553184 0.342019 1.514222]
inferred external torque RMS xyz [N m]: [0.049873 0.436937 0.08981]
legacy acceleration-residual loss diagnostic: 0.818567109454

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Parameter data objective

sample-summed uncentered squared dimensionless residual body wrench
force scale F^* = 23.0612945139 N
torque scale τ^* = 4.41819797713 N m
residual mean is NOT subtracted before optimization

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message='gtol' termination condition is satisfied.
reported optimality=8.836106690779388e-08
initial cost=6.510815988888972
final cost=6.510780249016224
cost reduction=3.5739872747519996e-05
coordinate step L2=0.00897143092107601
final gradient inf-norm=8.836106690779388e-08
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=552.667974385779
max directional finite-difference Jacobian relative error=7.267064508777685e-12